

Privacy-Preserving Natural Language Processing

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Lecture 2 — Text anonymization

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www.trusthlt.org

Trustworthy Human Language Technologies Group (TrustHLT)

Ruhr University Bochum & Research Center Trustworthy Data Science and Security



CENTER FOR TRUSTWORTHY
DATA SCIENCE AND SECURITY

Motivation

- 1 Motivation
- 2 Legal framework of HIPAA privacy rule
- 3 Anonymization and the GDPR
- 4 Case studies in anonymization
- 5 Text anonymization beyond short PII spans
- 6 Limitations of anonymization
- 7 Limitations of research on anonymization

How to protect information in text

The most straightforward and commonly used technique for protecting [REDACTED] in documents is [REDACTED].

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It has been used in the context of TASK_PLACEHOLDER_1, TASK_PLACEHOLDER_2, or even [REDACTED] since [REDACTED].

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This heavily redacted and altered paragraph bears little utility for learning something new about text anonymization, but at least clearly demonstrates how that technique works.

Trade-offs: Two assumptions

First, there is a need to publish a piece of text, say a classified secret document, a medical record, or a court judgment. Publishing such a document provides some non-zero utility to the public.

Second, the original document contains secrets that must be kept secret, or private information that must be kept private.

Legal framework of HIPAA privacy rule

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Anonymization in health domain: Usually deemed private

The “Health Insurance Portability and Accountability Act of 1996” is a public law in the U.S. which governs the privacy of health information, among other aspects, and explicitly defines the process of “de-identification, by which identifiers are removed from the health information,” with the aim to “mitigate privacy risks to individuals and thereby support the secondary use of data for comparative effectiveness studies, policy assessment, life sciences research, and other endeavors.”

Clear intended utility: health data published for the social good

De-identification in HIPAA

Paragraph §164.514/a defines the de-identification standard as follows

“Health information that does not identify an individual and with respect to which there is *no reasonable basis to believe that the information can be used to identify an individual* is not individually identifiable health information.”

If we believe we cannot identify an individual from the given piece of information (e.g., text), then this piece of information is considered safe for public release by HIPAA.

How should we implement it? HIPAA Safe Harbor

Lists of “identifiers of the individual or of relatives, employers, or household members of the individual” that must be removed (§164.514/a/2/i)

18 categories: names, all geographic subdivisions smaller than a state, such as street address, city, county, precinct, and zip code, all elements of dates (except year) for dates directly related to an individual, etc.

Last three categories fuzzy, e.g., category (P): biometric identifiers, including finger and voice prints, category (Q): full face photographic images and any comparable images, and category (R): any other unique identifying number, characteristic, or code.

Anonymization and the GDPR

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Personal data

Data is deemed personal if the information relates to an identified or identifiable individual

Art. 4 No. 1 GDPR.

General Data Protection Regulation — a European Union regulation on information privacy in the European Union (EU) and the European Economic Area (EEA)

Enhance individuals' control and rights over their personal information and to simplify the regulations for international business

P. Voigt and A. von dem Bussche (2017).
The EU General Data Protection Regulation (GDPR): A Practical Guide.
Springer International Publishing

Personal data

P. Voigt and A. von dem Bussche (2017).
The EU General Data Protection Regulation (GDPR): A Practical Guide.
Springer International Publishing

Individual does not need to be identified already.

The mere possibility of identification, **identifiability**, will render data **personal** under the GDPR

Identification — by combining different information that by themselves would not have traced back to the person but does so in combination

Personal data

“In October 2016, the ECJ ruled that the risk of identification appears insignificant in reality if it requires a disproportionate effort in terms of time, cost and manpower, the aforementioned being relative criteria.”

“Hence, a person can be considered as identifiable if the missing information that would allow identification is (easily) accessible, for instance, because it is published on the Internet or in a (commercial) information service.”

(Voigt and Bussche, 2017, p. 12)

P. Voigt and A. von dem Bussche (2017).
The EU General Data Protection Regulation (GDPR): A Practical Guide.
Springer International Publishing

Anonymization

A way of modification of personal data with the result that there remains no connection of data with an individual

Anonymised data is personal data that was rendered anonymous in such a manner that the person is no longer **identifiable**

“In case of an effective anonymisation, the GDPR does not apply”

(Voigt and Bussche, 2017, p. 13)

P. Voigt and A. von dem Bussche (2017).
The EU General Data Protection Regulation (GDPR): A Practical Guide.
Springer International Publishing

Anonymization: Practical Advice

As the EU does not provide for a standard of successful anonymisation, a combination of **randomisation** and **generalisation** techniques should be considered for stronger privacy guarantees

As a risk factor is always inherent to anonymisation, this must be considered when assessing possible techniques corresponding to the severity and likelihood of the identified risk. As a consequence, the optimal solution needs to be determined on a case-by-case basis

(Voigt and Bussche, 2017, p. 14)

P. Voigt and A. von dem Bussche (2017).
The EU General Data Protection Regulation (GDPR): A Practical Guide.
Springer International Publishing

Anonymization and the GDPR

Text anonymization in NLP

Text anonymization

Complete and irreversible removal of personally identifiable information (PII) from text data

PII

- Directly (name, passport, number, phone number, social media ID)
- Indirectly (gender, nationality, date of birth, etc.)

O. Yermilov, V. Raheja, and A. Chernodub (2023). **“Privacy- and Utility-Preserving NLP with Anonymized data: A case study of Pseudonymization”**. In: *Proceedings of the 3rd Workshop on Trustworthy Natural Language Processing (TrustNLP 2023)*. Toronto, Canada: Association for Computational Linguistics, pp. 232–241

Text anonymization

O. Yermilov, V. Raheja, and A. Chernodub (2023). **“Privacy- and Utility-Preserving NLP with Anonymized data: A case study of Pseudonymization”**.

In: *Proceedings of the 3rd Workshop on Trustworthy Natural Language Processing (TrustNLP 2023)*. Toronto, Canada: Association for Computational Linguistics, pp. 232–241

Is complete anonymization possible?

Two sub-tasks of anonymization

De-identification

Process of removing specific, predefined direct identifiers from a dataset

Pseudonymisation

Process of replacing direct identifiers with pseudonyms or coded values (such “John Doe” → “Patient 3”). The mapping between coded values and the original identifiers is then stored separately.

“NLP research on text anonymisation has focused to a large extent on the tasks of de-identification” (Lison, Pilán, Sanchez, Batet, and Øvrelid, 2021, p. 4190)

P. Lison, I. Pilán, D. Sanchez, M. Batet, and L. Øvrelid (2021). **“Anonymisation Models for Text Data: State of the art, Challenges and Future Directions”**. In: *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*. Online: Association for Computational Linguistics, pp. 4188–4203

Two sub-tasks of anonymization

De-identification generally modelled as a sequence labelling task, similar to Named Entity Recognition

P. Lison, I. Pilán, D. Sanchez, M. Batet, and L. Øvrelid (2021). **"Anonymisation Models for Text Data: State of the art, Challenges and Future Directions"**. In: *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*. Online: Association for Computational Linguistics, pp. 4188–4203

Case studies in anonymization

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Case-study: Job postings

- Task: detecting privacy-related entities in job posts on StackOverflow
- Intention: cannot identify which company posted the job (name of the employees who posted and their contact information)
- New data: 22k EN sentences annotated with person names, contact details, and profession
- Five entities (Org, Loc, Contact, Name, Profession)
- Annotation study: three annotators, good kappa, well-done study
- Models: entity taggers (LSTM, BERT) + auxiliary tasks

K. N. Jensen, M. Zhang, and B. Plank (2021). **"De-identification of Privacy-related Entities in Job Postings"**. In: *Proceedings of the 23rd Nordic Conference on Computational Linguistics (NoDaLiDa)*. Reykjavik, Iceland (Online): Linköping University Electronic Press, Sweden, pp. 210–221

Case-study: German e-mails

- Focuses on PII recognition and substitution with surrogates
- Dataset: "CodE Alltag", German e-mails from a) USENET and b) donations
- Entity types: more than NER - names, orgs, city names, zip codes, street names, street numbers, dates, passwords, e-mails, URLs, phone numbers (see the next slides)
- Experiments: detect PII (BIO tagging)

E. Eder, M. Wiegand, U. Krieg-Holz, and U. Hahn (2022). **“Beste Grüße, Maria Meyer” — Pseudonymization of Privacy-Sensitive Information in Emails**”. In: *Proceedings of the Thirteenth Language Resources and Evaluation Conference*. Marseille, France: European Language Resources Association, pp. 741–752

Case-study: German e-mails — PII types

<i>pi</i> Entity Type	Abbreviation
family names	FAMILY
female given names	FEMALE
male given names	MALE
organizations	ORG
user names	USER
city names	CITY
zip codes	ZIP
street names	STREET
street numbers	STREETNO
dates	DATE
passwords	PASS
unique formal identifiers	UFID
email addresses	EMAIL
phone numbers	PHONE
URLs	URL

E. Eder, M. Wiegand, U. Krieg-Holz, and U. Hahn (2022). **“Beste Grüße, Maria Meyer” — Pseudonymization of Privacy-Sensitive Information in Emails**”. In: *Proceedings of the Thirteenth Language Resources and Evaluation Conference*. Marseille, France: European Language Resources Association, pp. 741–752

Case-study: Geman e-mails — Overview



E. Eder, M. Wiegand, U. Krieg-Holz, and U. Hahn (2022). **“Beste Grüße, Maria Meyer” — Pseudonymization of Privacy-Sensitive Information in Emails**. In: *Proceedings of the Thirteenth Language Resources and Evaluation Conference*. Marseille, France: European Language Resources Association, pp. 741–752

Case-study: German e-mails — Evaluation

BIO scheme: 'B' — Beginning of an entity, 'I' (Inside) for its continuation, and 'O' (Outside) for tokens that do not belong to any pi entity)

Weighted average of precision, recall and F1 score

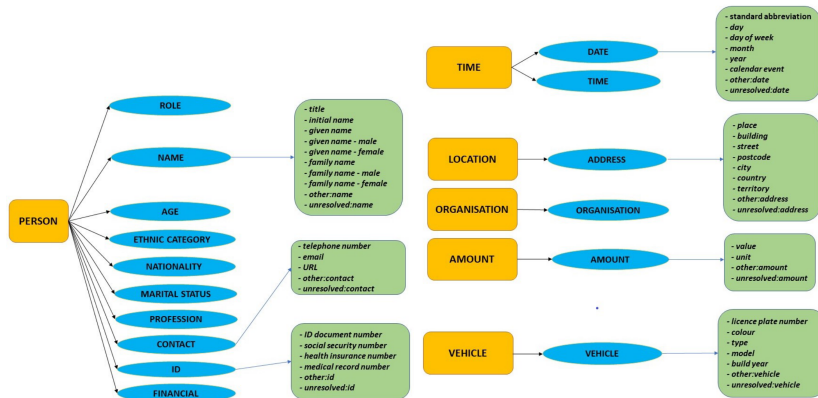
E. Eder, M. Wiegand, U. Krieg-Holz, and U. Hahn (2022). **“Beste Grüße, Maria Meyer” — Pseudonymization of Privacy-Sensitive Information in Emails**. In: *Proceedings of the Thirteenth Language Resources and Evaluation Conference*. Marseille, France: European Language Resources Association, pp. 741–752

Model Architecture	(Concatenated) Embeddings	Document Context	Prec	Rec	F ₁
Transformer	GELECTRA-LARGE	✓	87.14	87.64	87.31
Transformer	GELECTRA-LARGE + FASTTEXT + BPEMB		86.80	88.44	87.56
Transformer	GELECTRA-LARGE		87.73	87.58	87.52
Transformer	GELECTRA-LARGE + FASTTEXT + BPEMB		88.10	88.27	88.09

How robustly does the system protect privacy?

Case-study: MAPA project

- 3-level hierarchy of entities: level 1: person, time, location, organization, amount, vehicle



V. Arranz, K. Choukri, M. Cuadros, A. García Pablos, L. Gianola, C. Grouin, M. Herranz, P. Paroubek, and P. Zweigenbaum (2022). **“MAPA Project: Ready-to-Go Open-Source Datasets and Deep Learning Technology to Remove Identifying Information from Text Documents”**. In: *Proceedings of the Workshop on Ethical and Legal Issues in Human Language Technologies and Multilingual De-Identification of Sensitive Data In Language Resources within the 13th Language Resources and Evaluation Conference*. Marseille, France: European Language Resources Association, pp. 64–72

Case-study: MAPA project

- NER model: BERT
- Classification only in 2 levels of hierarchy
- Rntity replacement: random or using a LM
- Datasets: EUR-LEX (2k sentences per lang)
- Medical data: created “fake ones” in French by replacing “he”, “she”, “the patient” with fake names
- Also some legal judgment in Spanish
- Annotated data machine-translated

V. Arranz, K. Choukri, M. Cuadros, A. García Pablos, L. Gianola, C. Grouin, M. Herranz, P. Paroubek, and P. Zweigenbaum (2022). **“MAPA Project: Ready-to-Go Open-Source Datasets and Deep Learning Technology to Remove Identifying Information from Text Documents”**. In: *Proceedings of the Workshop on Ethical and Legal Issues in Human Language Technologies and Multilingual De-Identification of Sensitive Data In Language Resources within the 13th Language Resources and Evaluation Conference*. Marseille, France: European Language Resources Association, pp. 64–72

Case study: Pseudonymization with LLMs

- Pseudonymization adopted from Eder, Wiegand, Krieg-Holz, and Hahn (2022) — recognize and replace entities
- Tested 3 approaches: NER, seq2seq (enc-dec), LLM with zero shot
- 3 classes of entities: PER, LOC, ORG
- Spacy for NER; for LLM they use GPT-3 and ChatGPT
- Downstream test on pseudoanonym. data -> summarization (CNN daily mail, standard) and classification (IMDB, standard)
- Results: on summarization drop in BLEU (quite a big one), on sentiment very little (why?)

O. Yermilov, V. Raheja, and A. Chernodub (2023). **“Privacy- and Utility-Preserving NLP with Anonymized data: A case study of Pseudonymization”**. In: *Proceedings of the 3rd Workshop on Trustworthy Natural Language Processing (TrustNLP 2023)*. Toronto, Canada: Association for Computational Linguistics, pp. 232–241

Case study: TAB Project

1,268 En cases from ECHR (Introduction and Facts sections)

- PERSON Names of people, including nicknames/aliases, usernames, and initials.
- CODE Numbers and identification codes, such as social security numbers, phone numbers, passport numbers, or license plates.
- LOC Places and locations, such as cities, areas, countries, addresses, named infrastructures, etc.
- ORG Names of organizations, such as public and private companies, schools, universities, public institutions, prisons, healthcare institutions, non-governmental organizations, churches, etc.
- DEM Demographic attributes of a person, such as native language, descent, heritage, ethnicity, job titles, ranks, education, physical descriptions, diagnosis, birthmarks, ages.
- DATETIME Description of a specific date (e.g., *October 3, 2018*), time (e.g., *9:48 AM*), or duration (e.g., *18 years*).
- QUANTITY Description of a meaningful quantity, e.g., percentages or monetary values.
- MISC Every other type of personal information associated (directly or indirectly) to an individual and that does not belong to the categories above.

I. Pilán, P. Lison, L. Øvreliid, A. Papadopoulou, D. Sánchez, and M. Batet (2022). **“The Text Anonymization Benchmark (TAB): A Dedicated Corpus and Evaluation Framework for Text Anonymization”**. In: *Computational Linguistics* 48.4, pp. 1053–1101

Case study: TAB Project

PROCEDURE

1. The case originated in an application (no. 17582/04) against the Republic of Turkey lodged with the Court under Article 34 of the Convention for the Protection of Human Rights and Fundamental Freedoms ("the Convention") by a Turkish national, Mr Eyüp Kaya ("the applicant"), on 26 April 2004.
2. The applicant was represented by Mr M. Timur, a lawyer practising in Van. The Turkish Government ("the Government") were represented by their Agent.
3. On 18 September 2007 the Court decided to give notice of the application to the Government. It also decided to examine the merits of the application at the same time as its admissibility (Article 29 § 3).

THE FACTS

I. THE CIRCUMSTANCES OF THE CASE

1. The applicant was born in 1980 and lives in Van.
2. On 29 August 2000 the applicant was admitted to the military service.
3. On 7 September 2001 he went to see a doctor at the Sivas Military Hospital for an eye-sight problem. His medical report stated that he had - 6, 25 of myopia (nearsightedness), +0,25 of hyperopia (farsightedness) and - 6,75 of anisometropia amblyopia (difference in refractive error between the two eyes leading to reduced vision in one eye) on his left eye. The report noted that the applicant had complained that he had had the problem on his left eye since childhood. Accordingly, the applicant was discharged from the military as he was no longer eligible for service.

I. Pilán, P. Lison, L. Øvreliid, A. Papadopoulou, D. Sánchez, and M. Batet (2022). "The Text Anonymization Benchmark (TAB): A Dedicated Corpus and Evaluation Framework for Text Anonymization". In: *Computational Linguistics* 48.4, pp. 1053–1101

Case study: TAB Project — Evaluation

Evaluation - recall of PII = privacy protection, precision of PII = utility

Absolute recall — relies on a uniform weight over all annotated identifiers and thus fails to account for the fact that some (quasi-)identifiers have a much larger influence on the disclosure risk than others

Evaluation of anonymization methods is the need to compute the recall at the level of entities rather than mentions. Whenever an entity is mentioned several times in a given document, it only makes sense to view this entity as “protected” if all of its mentions are masked.

I. Pilán, P. Lison, L. Øvreliid, A. Papadopoulou, D. Sánchez, and M. Batet (2022). **“The Text Anonymization Benchmark (TAB): A Dedicated Corpus and Evaluation Framework for Text Anonymization”**. In: *Computational Linguistics* 48.4, pp. 1053–1101

Case study: TAB Project — New metrics

ANNOTATOR 1
ANNOTATOR 2
SYSTEM 1 (Longformer)
SYSTEM 2 (Presidio)

The case originated in an application (no. 12345/67) against

① (DIRECT)
① (DIRECT)

ANNOTATOR 1
ANNOTATOR 2
SYSTEM 1 (Longformer)
SYSTEM 2 (Presidio)

the Kingdom of Sweden by a British national, Mr John Doe,

② (QUASI) ② (QUASI) ③ (DIRECT)
③ (DIRECT)

ANNOTATOR 1
ANNOTATOR 2
SYSTEM 1 (Longformer)
SYSTEM 2 (Presidio)

on 1 October 2021. Mr Doe is a British researcher.

④ (QUASI) ③ (DIRECT) ② (QUASI) ⑤ (QUASI)
④ (QUASI) ③ (DIRECT)

I. Pilán, P. Lison, L. Øvrelid, A. Papadopoulou, D. Sánchez, and M. Batet (2022). **“The Text Anonymization Benchmark (TAB): A Dedicated Corpus and Evaluation Framework for Text Anonymization”**. In: *Computational Linguistics* 48.4, pp. 1053–1101

Entity is correctly masked if and only if the anonymization model manages to completely mask all of its mentions

Separate recall measures for the direct id (person's names, case numbers, etc.) and the quasi-id (dates, locations, etc.)

Text anonymization beyond short PII spans

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Information about others

So far we have relied on two main implicit assumptions:

- First, the PII text snippets to be redacted were usually very short, which corresponds to the notion of entities in NER
- Second, it was the information about a *third party* that was the anonymized. For example, in medical records we anonymize the patients, in the court decision we protect the victims, and so on—to make releasing the data compliant with the given legal framework

Confidentiality vs. self-disclosure

Revealing less about themselves (confidentiality) or revealing more about themselves (self-disclosure) are natural features of human day-to-day communication

Dou, Krsek, Naous, Kabra, Das, Ritter, and Xu (2024) detect self-disclosure phrases

"I am pretty sure I have autism, but I don't want to get an official diagnosis." labeled as Health (bold text), *"I'm a motorcycle tourer (by profession), but when I'm off the saddle I'm mostly bored"* labeled as Occupation, or *"Yes. I was making \$68k a year and had around \$19k in debt"* labeled as Finance.

Y. Dou, I. Krsek, T. Naous, A. Kabra, S. Das, A. Ritter, and W. Xu (2024). **"Reducing Privacy Risks in Online Self-Disclosures with Language Models"**. In: *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Ed. by L.-W. Ku, A. Martins, and V. Srikumar. Bangkok, Thailand: Association for Computational Linguistics, pp. 13732–13754

Inferences about personal attributes with LLMs

Protecting some private attributes of the author, such as a place of living, even if this information is not explicitly mentioned in the text but can be inferred from the text by a strong LLM, for example:

"This left shark thing is hilarious, I've seen it, walked just by there to see it live after my final exams."

This text alone allows an "LLM to infer that the author lived in Glendale, where the 2015 Super Bowl halftime show featured a performance colloquially called 'left shark incident'."

R. Staab, M. Vero, M. Balunovic, and M. Vechev (2025). **"Language Models are Advanced Anonymizers"**. In: *Proceedings of the Thirteenth International Conference on Learning Representations*. Singapore: Curran Associates, Inc

Limitations of anonymization

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Output

De-identified

Hello, my name is <PERSON> and I live in <LOCATION>.

My credit card number is <CREDIT_CARD> and my crypto wallet id is <ID>

<LOCATION> is a beautiful city in <LOCATION>!

We also visited <LOCATION>, the city of love! By the way, my e-mail password is "s!cret".

Presidio

Output

De-identified

Hello, my name is <PERSON> and I live in <LOCATION>.

My credit card number is <CREDIT_CARD> and my crypto wallet id is <ID>

<LOCATION> is a beautiful city in <LOCATION>!

We also visited <LOCATION>, the city of love! By the way, my e-mail password is "s!cret".

Input

Enter text

Hello, my name is David Johnson and I live in Maine.

My credit card number is 4095-2609-9393-4932 and my crypto wallet id is 16Yeky6GMjeNkAiNcBY7ZhrLoMSg1BoyZ.

Prague is a beautiful city in Europe!

We also visited Paris, the city of love! By the way, my e-mail password is "s!cret".

Limits of anonymization?

VG Arnsberg, Urteil vom 23.04.2014 - 9 K 900/13

Fundstelle

openJur 2019, 21479

Rkr: ☐ ” i

Tenor

- ¹ Die Klage wird abgewiesen.
- ² Der Kläger trägt die Kosten des Verfahrens.
- ³ Das Urteil ist wegen der Kosten vorläufig vollstreckbar. Der Kläger darf die Vollstreckung in Höhe von 110 % des aufgrund des Urteils zu vollstreckenden Betrags abwenden, wenn nicht die Beklagte vor der Vollstreckung Sicherheit in Höhe von 110 % des jeweils zu vollstreckenden Betrags leistet.

Tatbestand

- ⁴ Der Kläger ist bei der Beklagten als Fernstudent im Studiengang "Bachelor of Laws" eingeschrieben.
- ⁵ Unter dem 10. September 2012 beantragte der Kläger unter Vorlage verschiedener Studien- und Leistungsnachweise anderer Ausbildungseinrichtungen die Anrechnung von Prüfungsleistungen für das Studium "Bachelor of Laws". Zur Anrechnung auf das Propädeutikum (Modul-Nr. 55100) legte er eine Bescheinigung der Rheinischen Friedrich-Wilhelm-Universität C. vom 30. Juli 2003 über die Teilnahme an einer Arbeitsgemeinschaft zur Vorlesung "Bürgerliches Recht-Allgemeiner Teil" einschließlich einer Abschlussklausur vor. Wegen der Anrechnung auf die Module "Bürgerliches Recht I" (Modul Nr. 55101) und "Bürgerliches Recht II" (Modul Nr. 55103) sowie das Modul "Deutsches und Europäisches Verfassungsrecht" (Modul Nr. 55104) verwies er auf das Zwischenprüfungszeugnis der Universität C. vom 14. Dezember 2010. Dem

Attacks on anonymized documents

Explored reconstructing Swiss court decisions (cases from the federal court)

Auxiliary data: Created a dataset by manually linking court rulings and newspaper articles using keywords, e.g., “4A_375/2021”, “10 years in prison”, etc.

A. Nyffenegger, M. Stürmer, and J. Niklaus (2024). **“Anonymity at Risk? Assessing Re-Identification Capabilities of Large Language Models in Court Decisions”**. In: *Findings of the Association for Computational Linguistics: NAACL 2024*. Mexico City, Mexico: Association for Computational Linguistics, pp. 2433–2462

 A “Artist Y’s real name is Person X.”	 B “Website y.com belongs to Artist Y.”	 C “Website y.com was involved in court decision {file_number}.”
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Limitations of research on anonymization

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LLM as strong de-identifiers?

PERSON_FIRSTNAME_1 PERSON_LASTNAME_2 is a
NUMERIC_4 year old actress, born in LOCATION_2 but
raised in LOCATION_1. PRONOUN is most famously known
for playing PERSON_FIRSTNAME_2
PERSON_LASTNAME_1 in the
OTHER_IDENTIFYING_ATTRIBUTE_2 franchise. [...]
PRONOUN is very involved in activism and is a very
outspoken feminist.

(Table 5 from Patsakis and Lykousas (2023))

C. Patsakis and N. Lykousas (2023).

**"Man vs the machine in the struggle
for effective text anonymisation in the
age of large language models".** In: *Sci-
entific Reports* 13.1, p. 16026

Shameless plug: Conundrum of privacy research

Emma Watson is a 31 year old actress, born in Paris but raised in the UK. She is most famously known for playing Hermione Granger in the Harry Potter franchise. [...] she is very involved in activism and is a very outspoken feminist.

Shameless plug: Conundrum of privacy research

S. Ochs and I. Habernal (2026). **"The Conundrum of Trustworthy Research on Attacking Personally Identifiable Information Removal Techniques"**. In: *Computational Linguistics*

Emma Watson is a 31 year old actress, born in Paris but raised in the UK. She is most famously known for playing Hermione Granger in the Harry Potter franchise. [...] she is very involved in activism and is a very outspoken feminist.

Angelina Jolie is a 49 year old actress, born in Los Angeles but raised in the Palisades, New York. She is most famously known for playing Lara Croft in the Lara Croft franchise. [...] she is very involved in activism and is a very outspoken feminist.

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Credits

Ivan Habernal

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Partly inspired by lectures from Antti Honkela, Aurélien Bellet, Gautam Kamath